Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
2	*(a)*		Arrow at C going left (1) Arrow at C going up (1)	2			2		
	(b)		Use of the equation $\frac{kq}{r^2}$ (1) 52 500 000 **or** 39 400 000 correct (1) Applying Pythagoras $(52.5\text{M}^2 + 39.4\text{M}^2)^{0.5}$ (1) Method for obtaining angle correct e.g. $\tan^{-1}\left(\frac{525}{394}\right)$ (1) Correct answers 65.6 MV m^{-1} **unit mark** and 53 or 37[$^{\circ}$] or 0.93 or 0.64 [rad] (1)	1	1 1 1 1		5	5	
	(c)		Use of the equation $\frac{kq}{r}$ (1) $\frac{28}{8}$ - $\frac{21}{6}$ or equivalent seen to be zero (1)	1	1		2	2	
	(d)		Conservation of energy applied e.g. $eV = ½\, mv^2$ or E_{k} calculated to be 1.34×10^{-15} [J] (1) 8 300 [V] (1) Negative sign (1) i.e. 8 300 [V] gets 2 marks, (-8 300 [V] 3 marks)		3		3	2	
			Question 2 total	**4**	**8**	**0**	**12**	**9**	**0**

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